

EE 350

Test 3

Instructions: This is a Take Home Test, based on the honor system. You can use any reference material that you want, but **you should work alone**. Consulting with your classmates or any other individual is not allowed and will be considering cheating. In that case no points will be given for the whole exam. If you have any questions regarding the test, please consult only with me or any of the TAs. Show all your work but try to be as concise as possible.

**This Test is Due on or Before Thursday, April 9th, 10:30 pm.
No exceptions.**

Submit a single pdf file on Canvas with your answers (and Matlab code if any)

Fully Justify your answers.

Question 1 (30%):

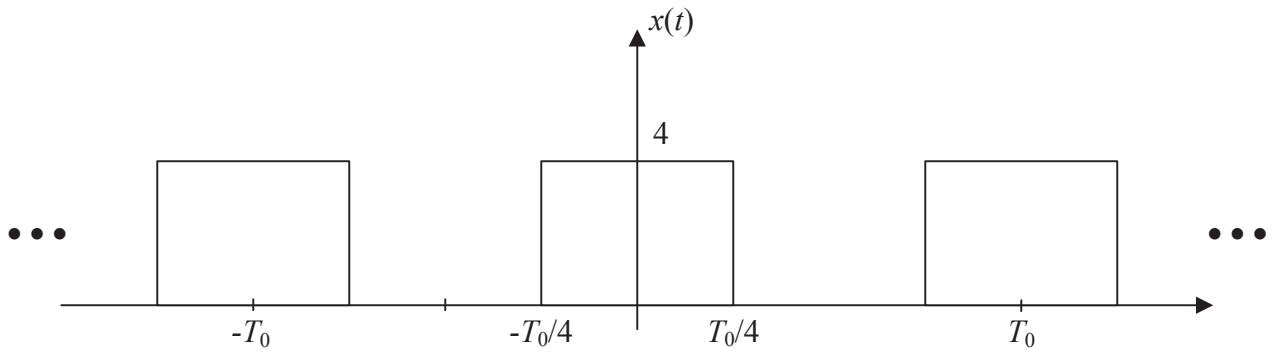
a) Consider the periodic signal

$$y(t) = \sin\left(2t + \frac{\pi}{4}\right) \sin\left(t + \frac{\pi}{3}\right) + 2 \cos^2(2t).$$

i) (10%) Specify the numeric value of the compact Fourier series parameters C_o , C_n , and θ_n .

ii) (8%) Using your results from part (a), determine the numeric values of the trigonometric Fourier series parameters a_o , a_n , and b_n .

b) (12%) Let $x(t)$ is the signal depicted below which is periodic with period T_0 .



Find the exponential Fourier series of $x(t)$

Question 2 (35%): The unit step response of an LTI system is:

$$y(t) = (-e^{-2t} + 1)u(t)$$

- a) (12%) Determine the ordinary differential equations that describes the system.
- b) (11%) What is the input $f(t)$ if the zero-state response $y_{zs}(t) = (2e^{-t} - 2e^{-2t})u(t)$.

Hint: If you have not done part (a), use differential equation

$$\frac{dy}{dt} + 2y(t) = 10f(t).$$

- c) (12%) Now consider an LTI system with the impulse response representation $h(t)$

$$h(t) = \frac{(t-3)^2}{2}e^{-2(t-3)}u(t-3).$$

Let the input to the system is given by

$$f(t) = \frac{1}{2\pi}e^{j4t}.$$

Using an appropriate transform pair and property from the table provided, determine the Fourier transform of the output $y(t)$?

Question 3 (35%): Throughout this question we aim at developing an easier way of implementing AM modulation.

a) (10%) Let $m(t)$ be the message to be sent and let

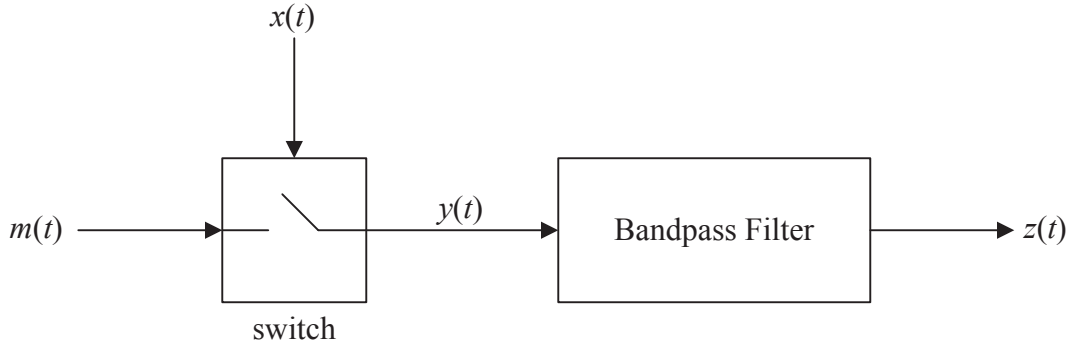
$$y_{mod}(t) = m(t) \cos(\omega_c t)$$

Determine the Fourier transform of $y_{mod}(t)$ as a function of the Fourier transform of $m(t)$.

b) (10%) Let $x(t)$ be the signal in part 1-b). Find the Fourier transform of $x(t)$.
If you have not done part 1-b) take

$$D_n = 3 \frac{\sin\left(\frac{n\pi}{3}\right)}{n\pi}$$

c) (15%) Because analog multiplication is difficult to implement over a wide dynamical range, in AM modulation sometimes the switching scheme below is used to obtain the modulated signal.



where $x(t)$ is the signal in part (b), $T_0 = \frac{2\pi}{\omega_c}$ and

$$y(t) = m(t)x(t).$$

Design the ideal bandpass filter above so that

$$z(t) = y_{mod}(t)$$

where $y_{mod}(t)$ is the signal defined in part (a).

If you have not done part (b) take

$$x(t) = \sum_{n=-\infty}^{+\infty} D_n e^{jn\omega_0 t} \quad \text{where} \quad \omega_0 = \frac{2\pi}{T_0}.$$

$$D_n = 3 \frac{\sin\left(\frac{n\pi}{3}\right)}{n\pi}$$